Investigating the effect of a natural botanical formula on the cognitive performance of rugby players: A within-subjects trial

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Summary:
This report aims to highlight the prevalence of concussions and brain injuries in English professional rugby and football, emphasising the link between traumatic brain injuries (TBI) and neurodegenerative diseases like chronic traumatic encephalopathy (CTE). More than that, it aims to understand the effects of the CONKA formula on Rugby Union players' cognitive performance.

The CONKA formula, developed by CONKA, focuses on natural ingredients known for their neuroprotective and cognitive enhancement properties. Key components include Ashwagandha, Turmeric with Black Pepper, Lemon Balm, Rhodiola Rosea, and Bilberry, each supported by scientific evidence for improving brain health and cognitive function in athletes.

A trial with 15 male rugby players from the Rugby Union Bristol Bears team evaluated the CONKA formula's effects on cognitive performance over 13 weeks. Cognitive tests showed a significant average improvement of 7.9% in focus, memory, information processing speed and accuracy among participants, suggesting the formula's efficacy in enhancing cognitive performance.

Analysis of Integrated Mouth Guard (IMG) data provided additional insights into physical impacts during gameplay, correlating with cognitive performance changes over time. However, technical issues affected data collection towards the end of the study, impacting results.

The study acknowledges limitations such as the absence of a control group and small sample size, recommending future research to validate findings and explore long-term effects. Despite these limitations, the findings support the notion that the CONKA formula can help mitigate cognitive decline associated with sports-related head traumas.

In conclusion, the report underscores the promising role of natural nootropics in supporting brain health and cognitive function in high-impact sports, contributing to ongoing research in sports medicine and brain injury prevention.

Background:

In English professional rugby, concussions account for 28% of injuries[1], while in football (soccer), they make up 22%[2]. This means that women playing football and rugby in England are 13 times more likely to suffer concussions and brain injuries than to be diagnosed with breast cancer annually, and men playing these sports are 39 times more likely to experience concussions and brain injuries than to be diagnosed with prostate cancer each year.[3]

Recent studies have indicated a link between traumatic brain injury (TBI) and neurodegenerative diseases[5, 6, 7, 8]. Specifically, chronic traumatic encephalopathy (CTE), a progressive neurodegenerative disorder, is believed to result from single or repetitive concussions and manifests as a range of motor, psychological, and cognitive symptoms[9, 10].

In rugby, repeated sub-concussive head traumas lead to a decline in cognitive performance and reduced blood flow to the brain after just one season, increasing the risk of long-term neurodegeneration[11]. In football, 80% of athletes are considered cognitively impaired after just 20 throw-in headers with a modern synthetic ball[12, 13]. Additionally, cognitive impairment is linked to a 60% risk of subsequent injuries (of any type) in Rugby Union[14] and a 50% risk in association football[15].

For individuals who have sustained two TBIs, there is a 33% increase in the risk of dementia, a risk that rises by 200% for those with five or more TBIs[16]. Despite government awareness and criticism of inaction[17], research on protecting and improving brain function remains limited. Therefore, this report aims to highlight the impact of nutrition on brain performance and test a formula developed by CONKA on professional Rugby Union athletes.

As Frances Arnold once said, "by far, nature is the best chemist of all time". With this in mind, the CONKA formula was developed using only natural ingredients that have pre-existing evidence to aid the maintenance of brain health and cognitive performance for athletes. This natural, food-based approach is especially relevant for elite athletes who must avoid substances prohibited by the World Anti-Doping Agency (WADA).

1. **Rationale for CONKA formula components**

Ashwagandha Root (Withania Somnifera): Anxiolytic; helps increase muscle mass and strength; anti-inflammatory; positive memory effects.

Turmeric (Curcuma Longa) with Black Pepper (Piper Nigrum): Anti-inflammatory; antioxidant; positive cardiac effects.

Lemon Balm (Melissa Officinalis): Calming effects; helps maintain attention; has anti-hyperexcitability and analgesic properties.

Rhodiola Rosea Root: Positive effects on fatigue (particularly mental fatigue); helps improve exercise performance (restorative); can improve mood; can improve memory performance.

Bilberry (Vaccinium Myrtillus): Anti-inflammatory; antioxidant; cardioprotective; positive effects on fatigue; antibiotic properties; can improve vision.

2. Evidence supporting this rationale

In addition to the detailed evidence supporting each component provided below, a breakdown of human trials conducted on each ingredient of the CONKA formula can be found in Supplementary Material 1.

2.1 Ashwagandha (Withania Somnifera)

Commonly called Ashwagandha, is a winter cherry tropical to the Solanaceae family. Ashwagandha contains various non-nutritional chemicals, responsible for its health-promoting properties.

Physical health effects:
Ashwagandha supplementation has been strongly linked to significant increases in muscle mass and strength, suggesting its potential usefulness when combined with a resistance-training program[18, 19]. Recent studies have also shown notable improvements in cardiorespiratory endurance such as VO_2max in both healthy adults and athletes with Ashwagandha supplementation[19, 20, 21].

Mental health effects:
Ashwagandha root extract has neuroprotective and neuro-reparative properties in neuronal cells, suggesting its therapeutic potential for addressing factors involved in secondary injury and long-term consequences of mild TBI. Studies using primary microglial cells and BV-2 microglial cell lines have shown that Ashwagandha root extract can inhibit microglial activation and migration, making it a potential candidate for suppressing neuroinflammation[22, 23]. In human studies, Ashwagandha root extract has recently been found to improve cognitive function[24], mood[25] and sleep quality[26].

2.2 Turmeric (Curcuma Longa) and Black Pepper (Piper Nigrum)

Physical health effects:
Dietary curcumin has been shown to improve endothelial function in humans[27]. It has also been proven to effectively inhibit maladaptive cardiac repair and preserve cardiac function after ischemia and reperfusion. Daily treatment with curcumin during reperfusion protects against maladaptive tissue repair and enhances cardiac function following ischemia. The cardioprotective effects of curcumin are linked to attenuation of lipid peroxidation and active MMPs, inhibition of the TGF1-Smad signalling pathway and differentiation of fibroblasts and modulations in the balance between collagen degradation and synthesis, all of which are associated with fibrosis. Through these beneficial effects, Turmeric helps maintain extracellular matrix (ECM) integrity and stability, crucial for cardiac recovery[28].

Mental health effects:
In humans, Turmeric has been found to improve many areas of cognitive function, including memory and attention[29, 30] but also mood and even the ability to decrease tau accumulation[31] (associated with cognitive problems). Additionally, several reports suggest that the well-established anti-inflammatory, antioxidant, and anti-apoptotic properties of Turmeric may be beneficial in treating brain injuries.

It has been observed that Turmeric can activate the expression of thioredoxin, an antioxidant protein in the Nrf2 pathway, which protects neurons from death caused by oxygen-glucose deprivation in an in vitro model of ischemia and reperfusion[32]. Furthermore, animal models have demonstrated the positive biochemical and morphological effects of Turmeric on the prefrontal cortex (PFC) and hippocampus, which are brain regions affected by chronic stress and crucial for learning and memory performance.

High levels of corticosterone in the hippocampus and PFC due to chronic stress cause atrophy of the dendrites of the key neurons and dendritic structural plasticity. Recent studies have shown that Turmeric not only increases brain-derived neurotrophic factor (BDNF) levels in the hippocampus and PFC[33, 34, 35], but also reduces serum corticosterone levels in animal models of chronic stress. Therefore, Turmeric can be considered a useful tool for improving synaptic plasticity and encouraging neural repair in the injured brain.

Since black pepper has been found to increase turmeric absorption by up to 2000%[52], it has also been included alongside turmeric in the CONKA product. Black pepper contains piperine, a natural compound that significantly enhances the absorption and bioavailability of curcumin, the active ingredient in turmeric. Curcumin has low bioavailability on its own, meaning that the body does not easily absorb it. However, piperine increases the absorption of curcumin by inhibiting certain digestive enzymes, thereby allowing more curcumin to enter the bloodstream and stay in the body longer. Combining turmeric with black pepper can thus enhance its effectiveness and potential health benefits.[51]

2.3 Lemon Balm (Melissa Officinalis)

Mental health effects:
Lemon Balm (LB) shows strong evidence for acute anxiolytic (anti-anxiety) effects as well as long-term anti-agitation effects. Human studies indicate that MO can enhance self-rated 'calmness' using established mood scales. Moreover, in controlled settings where subjects are exposed to stressors, LB has been observed to increase calmness[36, 37, 38]. The calming effects are believed to involve the suppression of hyperactive voltage-gated sodium channels[38].

Additionally, LB is noted for its potential to improve attention and cognitive performance[39, 40]. These mechanisms are also thought to contribute to its analgesic properties (Chazot et al., unpublished). LB is well-known for its cholinergic properties and there is evidence supporting its role in memory enhancement[41].

2.4 Rhodiola Rosea

Physical health effects:
Despite limited research exploring the effects of Rhodiola Rosea (RR) on physical health, evidence exists to support the notion that it can improve levels of physical fitness[42].

Mental health effects:
Repeated administration of RR extract has been shown to exert anti-fatigue effects, leading

to reduced cortisol, increased mental performance, and the ability to concentrate[43, 44]. Moreover, RR extract was found to be a safe and effective way of improving life-stress symptoms to a clinically relevant degree[45] when applied for a 4-week period, despite effects being observed after only 3 days of treatment.

In an interesting study looking at stress during a student exam period, RR produced significant improvement in a series of physical fitness, mental fatigue, and neuro-motoric tests[42]. In a cross-over study, healthy physicians on night duty displayed a statistically significant improvement in general fatigue observed in the treatment group (RRE) during the first two weeks period. The tests chosen reflect an overall level of mental fatigue, involving complex perceptive and cognitive cerebral functions, such as associative thinking, short-term memory, calculation and ability of concentration, and speed of audio-visual perception. Again, no side-effects were reported[43].

2.5 Bilberry (Vaccinium Myrtillus)

Physical health effects:
Bilberry (Vaccinium myrtillus L.) is among the richest natural sources of anthocyanins. It exhibits significant anti-inflammatory and lipid-lowering properties and has been shown to reduce oxidative stress, particularly in peripheral tissues. This reflects bilberry's potential for treating or preventing inflammation-related conditions, including both brain and peripheral injuries. In a study with mice subjected to restraint stress, bilberry extract restored elevated liver damage and reactive oxygen species (ROS) levels to normal, while also enhancing mitochondrial function[46].

Moreover, mice treated with Vaccinium myrtillus L. showed significant improvements in forelimb grip strength, prolonged weight-loaded swimming time, and reductions in lactate, ammonia, BUN, and CK activity. Additionally, their muscle and liver glucose and glycogen content increased compared to the control group. These findings indicate that plants rich in anthocyanins may possess anti-fatigue properties and enhance exercise tolerance. Interestingly, bilberry supplementation also reduced levels of soluble A40 and A42, compared to blackcurrant-fed mice, suggesting its potential as a protective measure against long-term neurodegenerative diseases[47].

Despite there being a lack of research conducted on humans with Bilberry, it is worth noting that trans-resveratrol has been found in Bilberry[48], which has previously been associated with an increase in tau phosphorylation in both rats[49] and human cells[50].

Methods:

To understand the effects of the CONKA product (see Table 1 for constituents) on rugby players, 15 male players from the Rugby Union Bristol Bears team performed baseline cognitive tests three times a week for one week before starting the intervention, then again three times a week for 12 weeks. Below are the constituents of the CONKA product (Table 1) and participants' information (Table 2). No participants were excluded or dropped out of the trial. The average compliance percentage was 66%. This calculation excludes the period after

March 20th due to technical difficulties with the cognICA test in the CONKA app, which prevented some players from testing regularly.

Table 1: Constituents of the CONKA product

Common name	**Scientific name**
Ashwagandha root	Withania somnifera
Turmeric with black pepper	Curcuma longa with piper negrum
Lemon balm	Melissa officinalis
Rhodiola rosea root	Rhodiola rosea
Bilberry	Vaccinium myrtillus

Table 2: Participant characteristics

Characteristic	**Total *n* = 15**
Age (yr)	26.80 ± 3.65
Height (m)	1.85 ± 0.06
Mass (kg)	104.40 ± 13.95
Playing position	9 Forwards, 6 Backs

Data is presented as mean ± standard deviation. Yr = Years; m = Metres; kg = Kilograms.

The cognitive tests included a variation of the Sports Assessment Concussion Tool (SCAT6) concentration test, a memory test and the Integrated Cognitive Assessment (CognICA). The CognICA test is an FDA and CE-approved computerised cognitive test based on a rapid categorisation task and is independent of language. It has been shown to accurately detect mild cognitive impairment and is moderately to highly correlated with other popular cognitive tests such as the Montreal Cognitive Assessment (Pearson's r = 0.58) and the Addenbrooke's Cognitive Examination (Pearson's r = 0.62)[53]. The ICA test provides three variables: ICA Speed (information processing speed), ICA Accuracy (information processing accuracy), and ICA Index (overall information processing ability, combining ICA Speed and ICA Accuracy), referred to as CognICA throughout this report.

During this 13-week trial, players were also taking an additional supplement including Omega 3 Fatty Acids (2464mg daily serving), EPA (Eicosapentaenoic acid; 1408mg), DHA (Docosahexaenoic acid; 704mg) and Astaxanthin (1000mg). In addition, 13 players wore an Integrated Mouth Guard (IMG) during matches and training sessions, which provided the IMG data available in Supplementary Material 2. This included average peak linear acceleration (PLA; g), peak angular acceleration (PAA; rad/sec^2), peak linear velocity (PLV; m/sec), peak angular velocity (PAV; rad/sec) and workload (J).

Results:

An average increase of +7.9% was observed across players on the 3 tests employed CognICA (ICA index) (which included 2 metrics ICA speed and ICA accuracy), focus, and memory scores when compared to baseline. This included an average increase of +19.3% in focus score, +2.6% in memory test and an increase of +7.9% in CognICA score which includes

+6.1% Speed and +3.9% accuracy. This is a sum total improvement across the CognICA, memory and focus scores, an increase of +33.2%.

Paired t-tests were conducted to assess significance, revealing significant results for all tests. The alpha *p* value was set to 0.05 (see Table 3). The scores of one player, who began the trial one week later, were excluded from the paired t-test calculations due to the potential for anomalies.

Table 3: Average cognitive score differences and p-values across all players

Cognitive test	**Baseline**	**Intervention**	**Change**	**Paired t-test (*p*-value)**
CognICA	75.3	81.3	6.1	0.009
ICA Speed	93.6	97.5	3.9	0.000
ICA Accuracy	80.9	83.6	2.6	0.523
Focus	65.4	84.7	19.3	0.000
Memory	74.2	82.1	7.9	0.009

Over the course of three-week intervals, the observed cognitive scores exhibited a gradual increase. Starting from a baseline of 77.9, scores rose to 83.9 at the first three-week interval, 86.5 at the second interval, and peaked at 88.7 at the third interval before slightly declining to 86.0 at the fourth interval.

Analysis of IMG data revealed a common pattern across all tests except for Peak Angular Velocity (PAV). Scores started lower at baseline, peaked during the first intervention interval (weeks 1 to 3), and dropped to below baseline levels in the second intervention interval (weeks 4 to 6). In contrast, PAV scores were highest at baseline, decreased slightly during the first interval, and decreased further in the second interval (see Table 4). IMG data was only collected until March 1st (interval 2).

Table 4: Average IMG data results across all players

Test	**Baseline**	**Interval 1**	**Interval 2**
Average Peak Linear Acceleration (PLA, g)	13.3	15.4	13.1
Peak Angular Acceleration (PAA, rad/sec^2)	983.1	1081.9	788.8
Peak Angular Acceleration (PAA, rad/sec^2)	1.0	1.1	0.9
Peak Angular Velocity (PAV, rad/sec)	9.2	8.5	6.4
Workload (J)	3.8	4.4	2.9

Discussion:

The findings of this trial highlight the benefits of the CONKA formula in enhancing cognitive performance among professional Rugby Union athletes. The results demonstrate significant improvements in all cognitive scores, including the FDA and CE-approved accuracy of processing information metric (CognICA), deeming the natural ingredients in the CONKA formula effective in supporting brain health and cognitive function.

The CONKA formula consists of Ashwagandha, Turmeric with Black Pepper, Lemon Balm, Rhodiola Rosea, and Bilberry, all of which have shown promise in supporting cognitive and physical health. Ashwagandha's neuroprotective and neuro-reparative properties, combined with its positive effects on mood and sleep which certainly promote recovery, may have contributed to the observed cognitive enhancements as these findings are in line with previous research[22, 24, 26]. Turmeric's anti-inflammatory and antioxidant effects, coupled with Black Pepper's role in enhancing curcumin absorption, likely played a role in improving cognitive function and reducing neuroinflammation, as well as improving attention and memory[29, 30]. Lemon Balm's anxiolytic effects and ability to enhance attention, along with Rhodiola Rosea's anti-fatigue and recovery-boosting properties, further support the overall cognitive benefits observed in this trial and especially the significant improvements in focus and cognitive processing measured by the cognICA test. These findings are also in line with previous research[39, 43, 44].

Cognitive Performance Improvements

The observed average increase of 7.9% across all tests, where significant increases include focus (19.3%), memory (7.9%), CognICA (6.1%), ICA speed (3.9%), and ICA accuracy (2.6%) scores, indicate that the CONKA formula positively impacts various aspects of cognitive performance. The significant p-values ($p \leq 0.05$) across all tests including focus ($p < .001$), memory ($p < 0.05$), CognICA ($p < 0.05$), ICA speed ($p < .001$) and ICA accuracy ($p \leq 0.05$), reinforce the efficacy of the formula in enhancing cognitive functions. These improvements are particularly noteworthy given the high incidence of cognitive impairments and long-term neurodegenerative risks associated with repetitive head traumas in contact sports like rugby and football[1,2].

The gradual increase in cognitive scores over the three-week intervals, peaking at the third interval before a slight decline, suggests a sustained benefit from the CONKA formula over time. This pattern may indicate an initial adaptation period (cognitive performance gradually increased from a baseline of 77.9 to 83.9 in the first interval, and further improved to 86.5 in the second interval), followed by an optimisation of cognitive performance (increasing to an 88.7 peak at the third interval), with potential factors such as compliance issues due to technical problems influencing the slight decline observed at the fourth interval (86.0).

The notion that cognitive performance improves gradually over time is supported by previous research on individual ingredients demonstrating similar effects: When observing the effects of Ashwagandha on sleep and anxiety, improvements were evident after 5 weeks, with even greater benefits after 10 weeks[60]. Similarly, research on Curcumin has observed a gradual improvement in both physical and mental health[61]. Additionally, previous studies on the CONKA formula's ingredients demonstrating benefits over periods longer than 9 weeks (length of 3 intervals), suggest there are no disadvantages to taking this product for extended durations[60, 61, 30]. Therefore, the slight decline observed at interval 4 may simply be a result of reduced compliance.

An alternative interpretation of this pattern may stem from a reduction in games played during intervals 2 and 3, a period where players may have received less impact (workload decreased from 4.4 at the first interval to 2.9 at the second interval; see Table 4) and therefore improved cognitively. However, it is important to note that substantial impact extends beyond games, as athletes also experience high levels of impact during contact training[54]. This underscores the relevance of this analysis both within the competitive season and during off-season periods. Moreover, given that IMG data is available only up to March 1st (interval 2), conclusive insights regarding the correlation between cognitive performance and contact exposure during intervals 3 and 4 cannot be derived.

Limitations and Future Directions

One limitation of this study is the lack of a control group. Although the inclusion of a control group is certainly crucial for more rigorous comparisons and internal validity, it is important to note that within-subject designs are also statistically powerful[55]. One other limitation of this trial involves the technical difficulties experienced after March 20th, which affected compliance rates and resulted in a lack of data at interval 4. Consequently, we were unable to conclude if the benefits of CONKA continued beyond the 9–12-week point. This highlights the need for robust and reliable testing methods. Future research should aim to include a larger sample size and a control group to obtain more accurate and representative results. Additionally, the long-term effects of the CONKA formula should be further explored using reliable tools to enhance the quality of the results and insights gathered. Additional investigation on the individual contributions of each component and potential synergistic effects would also provide valuable insights.

Conclusion

In conclusion, this study provides promising evidence for the cognitive benefits of the CONKA formula in professional rugby players. The significant improvements in cognitive scores underscore the potential of this natural, food-based formula in supporting brain health and cognitive function, particularly in high-risk groups such as contact sports athletes. These findings contribute to the growing body of research on natural nootropics and their role in enhancing cognitive performance and mitigating the risks associated with traumatic brain injuries.

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